

CHITRANG POTDAR

Backend Engineer | Applied AI

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PROFESSIONAL SUMMARY

Backend Engineer specializing in Python systems and Applied AI. Experienced designing REST APIs, autonomous workflows, multi-tenant architectures, and AI-enabled services with focus on maintainability and reliability. Built developer tooling, RAG pipelines, automation systems, and cloud-native applications using modern Python technologies. Delivered systems spanning investment platforms, AI pipelines, LMS/CMS, and full-stack productivity software.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Backend: FastAPI, Flask, REST APIs, Celery

Databases & Caching: PostgreSQL, Redis, MSSQL

AI & Agent Frameworks: Claude API, LlamaIndex, LangChain, LangGraph, MCP, AWS Transcribe, Google Cloud STT, Whisper

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD

Third-Party Integrations: SendGrid, Vimeo, Web Push (VAPID)

Observability: Grafana, Loki, Promtail, LangSmith

Testing & Automation: Pytest, Playwright

ML & Data: Scikit-learn, Pandas, NumPy, Polars, Seaborn, Matplotlib

PROFESSIONAL EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd., Pune

Oct 2025 – Present

Investment Operations Platform

Flask · PostgreSQL · Redis · Docker · GitHub Actions · AWS EC2 · Grafana · Loki · Promtail · Pytest · Playwright

- Built a multi-tenant backend with query-level data isolation across users, portfolios, and organizations — ensuring each tenant only ever accesses their own data within a shared PostgreSQL schema.
- Developed REST APIs for portfolio review, rejected order tracking, and financial calculators; integrated the Jira API to automatically create issues from application events, eliminating a manual reporting step.
- Reduced redundant database round-trips by introducing Redis-based session caching for frequently accessed user context; maintained 80%+ unit test coverage with Pytest.
- Automated NSE rejected order processing using Python API integrations and end-to-end client workflows using Playwright, reducing manual operational effort by approximately 2–3 hours per week.
- Configured Promtail to ship application logs into Loki and built Grafana dashboards to surface service health and error trends — giving the team structured visibility into backend behaviour for the first time.
- Built a domain-specific MCP server to expose investment platform data to LLM clients: annotated each tool and function signature with investment-domain terminology (e.g. portfolio review, order status) so the model correctly resolves business concepts to the underlying database operations without requiring raw schema knowledge.
- Implemented async background email delivery using Redis and Celery: on task creation, a job is enqueued in Redis, a Celery worker picks it up and sends the email, and the UI receives a failure notification if delivery does not succeed.
- Added Web Push notifications (VAPID-based) for single-user real-time alerts, extending the notification system beyond email.

AI Meeting Intelligence Pipeline

AWS Lambda · API Gateway · S3 · AWS Transcribe · Claude API

- Designed an async serverless pipeline to work within API Gateway's 29-second timeout: audio and agenda files are uploaded to S3, transcribed via AWS Transcribe, then passed to the Claude API to produce structured output — summary, agenda scoring, and action items.

AI Voice Conversational System with RAG

FastAPI · Google Cloud STT / Whisper · TTS · Claude API · LlamaIndex

- Built a voice AI assistant pipeline: speech-to-text → RAG retrieval (semantic chunking + embedding-based lookup) → LLM response → text-to-speech, grounding answers in a domain knowledge base.
- Designed the pipeline modularly so the knowledge base can be updated independently without changes to the core processing logic.

LMS & CMS Platform — Jaro Education

FastAPI · PostgreSQL · SendGrid · Vimeo · AWS S3 · Modular Monolithic Architecture

- Designed and built a modular monolithic backend for an LMS and CMS platform.
- Implemented authentication with email verification via SendGrid.
- Built course creation and student management modules for platform administrators.
- Integrated Vimeo for course video upload and streaming, and AWS S3 for certificate storage and retrieval through backend APIs.
- Developed backend APIs supporting the student-facing LMS for course access and certificate retrieval.

Conference Management CMS — IISc Bangalore

FastAPI · PostgreSQL

- Built a conference management system supporting conference publishing, participant registration, admin approval/rejection workflows, and attendance tracking.

Python Developer Intern · ArrayPointer Pvt. Ltd., Pune

Dec 2023 – Apr 2024

- Automated structured data extraction from PDFs (electricity bills, CAS files) using PyPDF2, Camelot, and Tabula, reducing manual extraction effort by approximately 70% with JSON output.
- Built an XGBoost revenue forecasting model with incremental learning support, improving forecast accuracy by roughly 20% as new data was ingested.

PROJECTS

SOLID-Agent-Harness — Multi-Agent Code Analysis Tool

Node.js · LangChain · LangGraph | github.com/Dev-Chitrang/SOLID-Agent-Harness

- Built a language-agnostic CLI developer-tooling platform running deterministic, graph-based LLM workflows for repository analysis, documentation generation, and code quality — six individual commands (audit, bugs, docs, readme, review, explain) plus five composite workflows.
- Implemented a critic-loop pattern per graph (Agent → Critic → conditional retry) that exits on a quality score threshold or iteration cap, improving output reliability over a single-pass agent call.
- Designed composite commands using LangGraph's parallel fan-out with a barrier-join summary node, running multiple analysis graphs concurrently and synthesizing results into one executive report.
- Built a provider-abstraction layer supporting OpenAI, Anthropic, Gemini, and NVIDIA, swappable without changing any graph code, plus an interactive TUI for provider/command/model selection.
- Architected registry-driven command dispatch so new commands can be added via a folder and registry entry alone, with no changes to core harness or TUI files.

Unified Workspace — Full-Stack Productivity Suite

FastAPI · React · PostgreSQL · Redis · Celery · Docker | unified-workspace.duckdns.org | github.com/Dev-Chitrang/Productivity-SaaS-Ful-Stack-App

- Built and deployed a self-hosted productivity platform unifying task management, notes, calendar scheduling, WebRTC audio meetings, collaborative whiteboarding, and AI meeting analysis into one application.
- Designed a layered Repository → Service → Controller architecture with dependency injection throughout the FastAPI backend for testability and separation of concerns.
- Built a Redis-backed meeting engine with session-scoped lifecycle management and WebSocket signaling for real-time audio meetings.
- Integrated AI-powered meeting analysis via NVIDIA NIM (Llama 3.3 70B) to generate summaries, agenda coverage, and actionable task suggestions from meeting transcripts.
- Implemented JWT authentication with Redis-backed token revocation, OTP-based two-factor auth, and Google OAuth sign-in.
- Maintained 100+ test files with 2,100+ test cases using pytest-asyncio; deployed as a 6-service Dockerized stack, live at unified-workspace.duckdns.org.

EDUCATION

B.E. Computer Engineering — 8.65 CGPA

Savitribai Phule Pune University, Pune | 2021 – 2025

CERTIFICATIONS

- IBM AI & Cloud — AICTE Certified
- VOIS AI & Data Analysis — AICTE Certified

RESEARCH & PUBLICATIONS

Corporate Credit Rating Analysis using AI & Sentiment Analysis

IJIRCCCE | Impact Factor: 8.65 | Nov 2024 | github.com/Dev-Chitrang/Final-year-project/tree/Backend